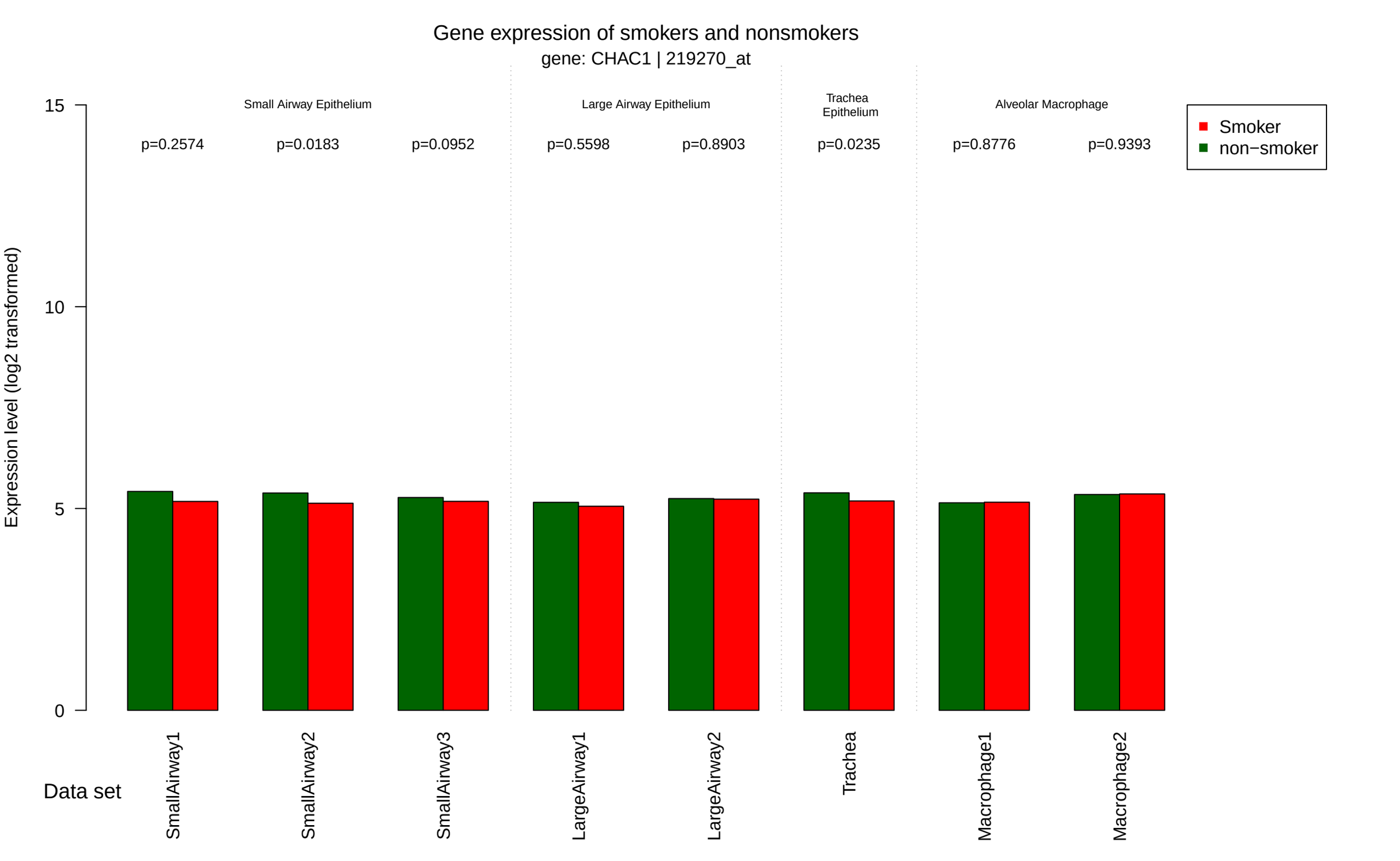
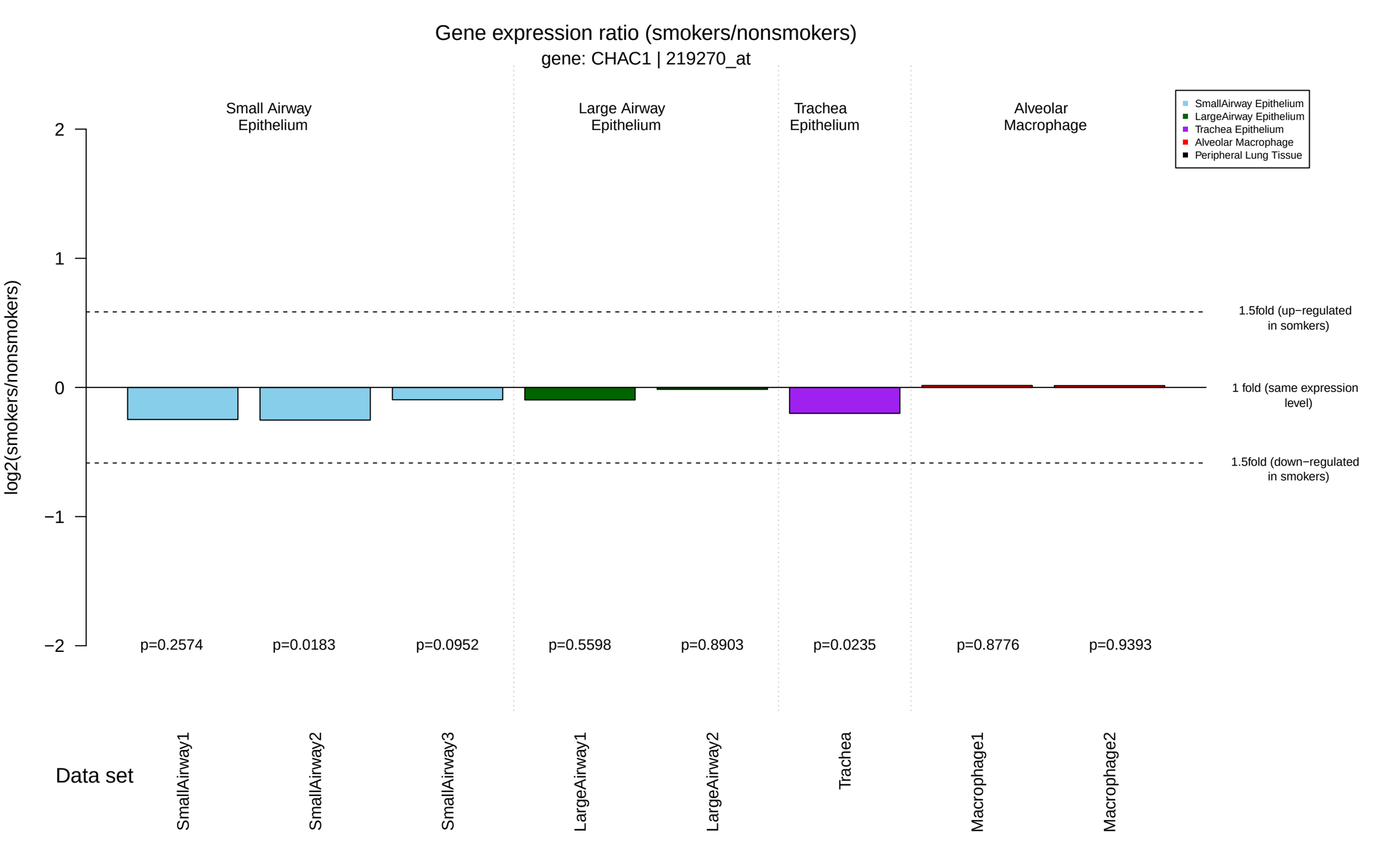
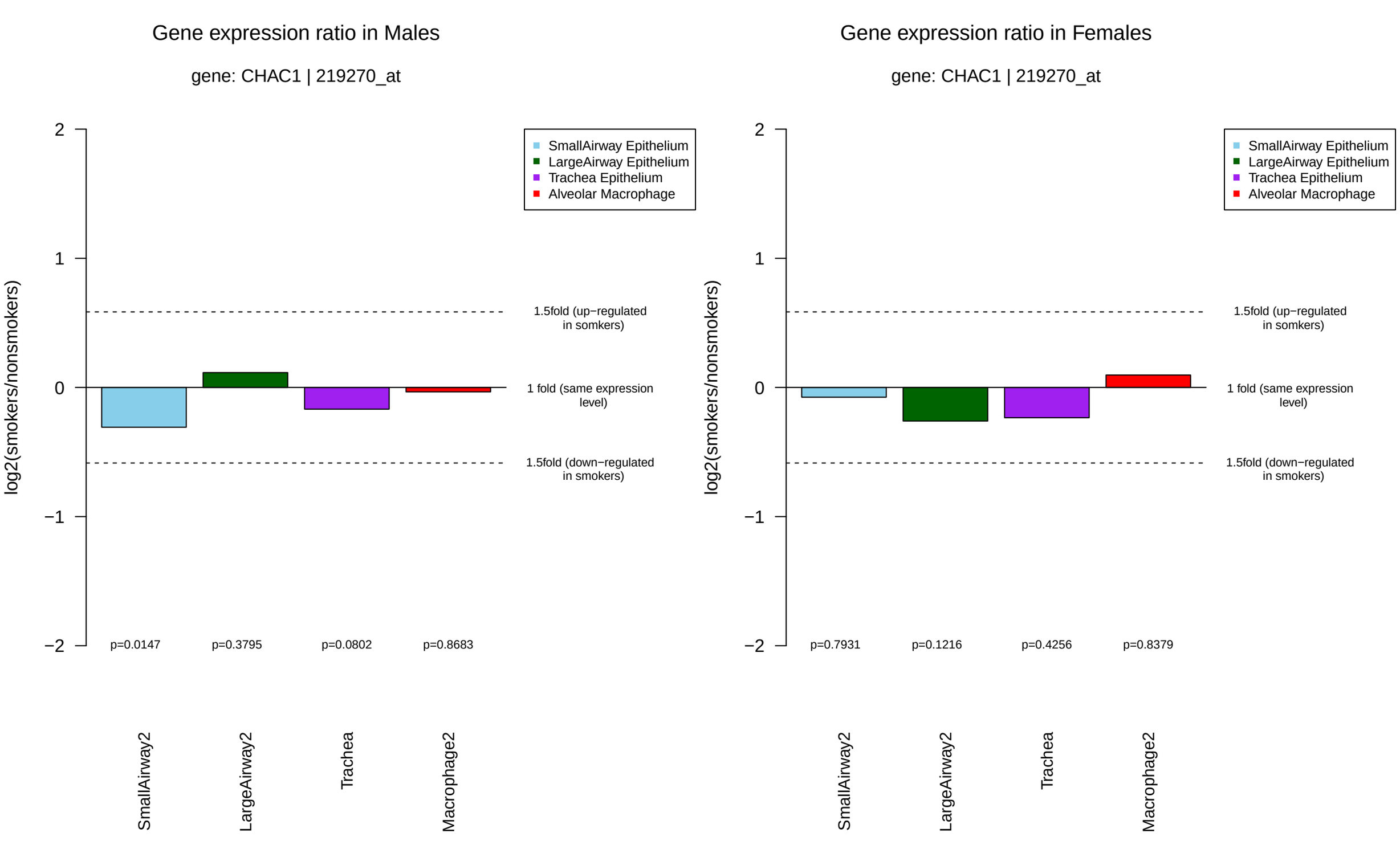
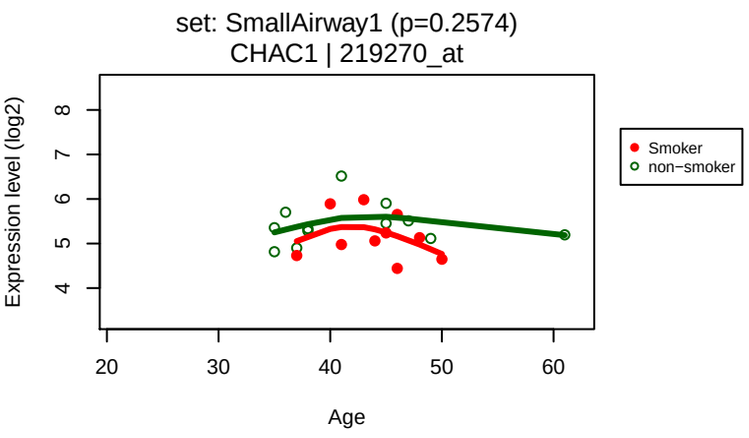
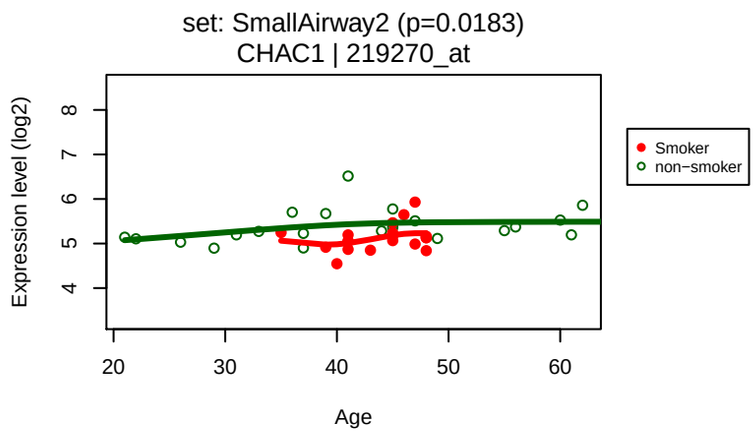
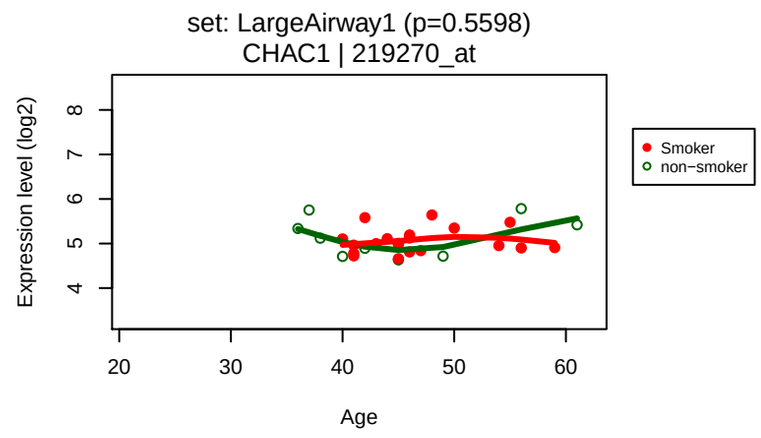

Gene expression of smokers and nonsmokers
gene: CHAC1 | 219270_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2574
p=0.0183
p=0.0952
p=0.5598
p=0.8903
p=0.0235
p=0.8776
p=0.9393
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

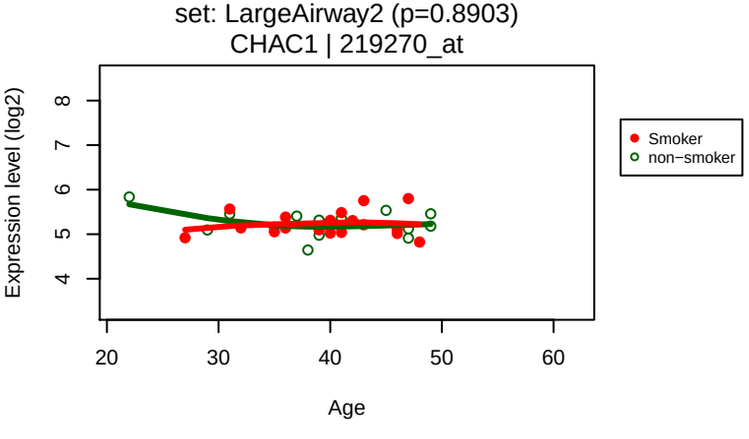

Gene expression ratio (smokers/nonsmokers)
gene: CHAC1 | 219270_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2574
p=0.0183
p=0.0952
p=0.5598
p=0.8903
p=0.0235
p=0.8776
p=0.9393
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

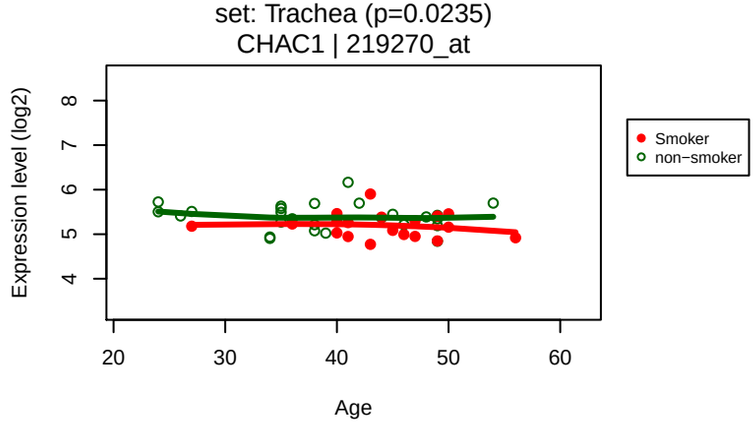

Gene expression ratio in Males
gene: CHAC1 | 219270_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0147
p=0.3795
p=0.0802
p=0.8683
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHAC1 | 219270_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7931
p=0.1216
p=0.4256
p=0.8379
SmallAirway2
LargeAirway2
Trachea
Macrophage2

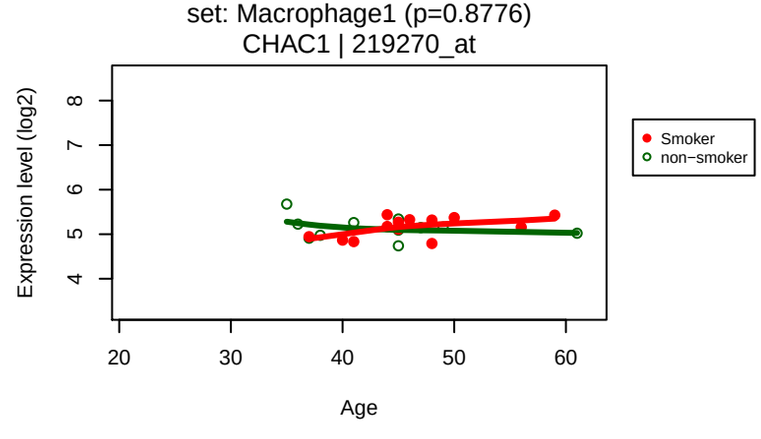

set: SmallAirway1 (p=0.2574)
CHAC1 | 219270_at
Expression level (log2)
Age
Smoker
non−smoker

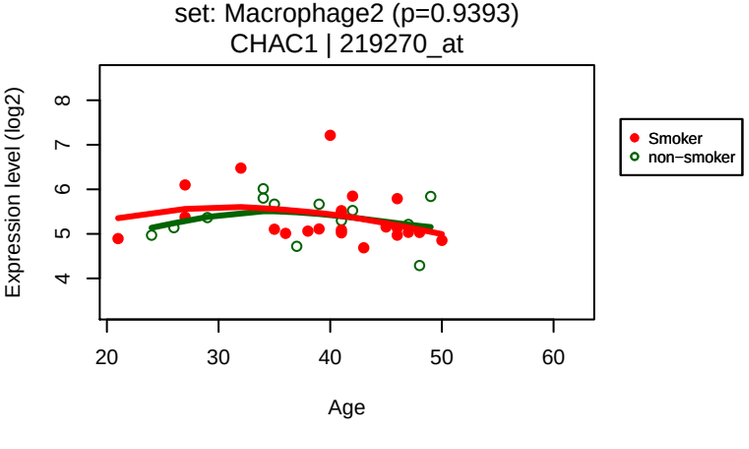

set: SmallAirway2 (p=0.0183)
CHAC1 | 219270_at
Expression level (log2)
Age
Smoker
non−smoker

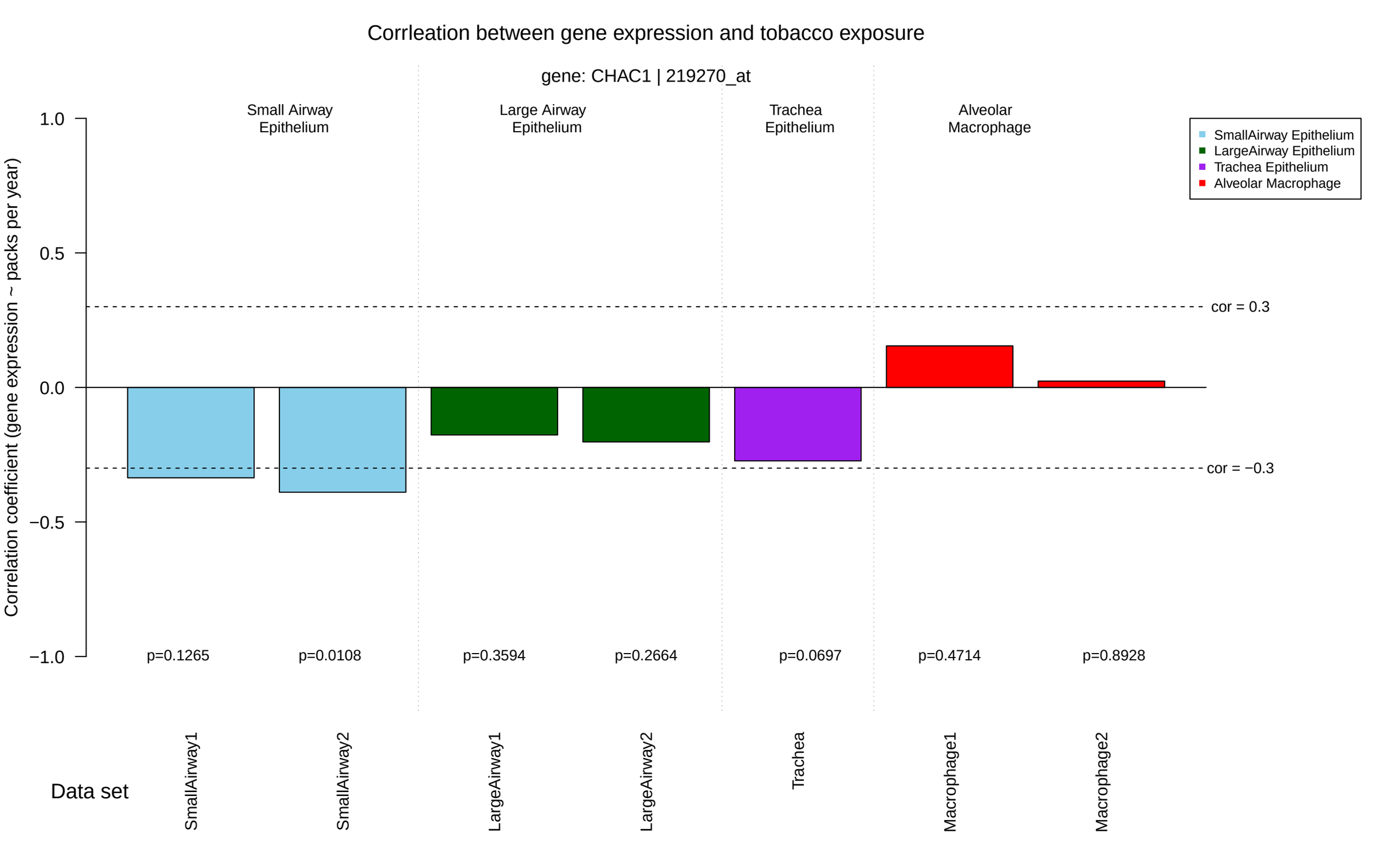

set: LargeAirway1 (p=0.5598)
CHAC1 | 219270_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8903)
CHAC1 | 219270_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0235)
CHAC1 | 219270_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8776)
CHAC1 | 219270_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9393)
CHAC1 | 219270_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHAC1 | 219270_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1265
p=0.0108
p=0.3594
p=0.2664
p=0.0697
p=0.4714
p=0.8928
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2